

CELL TRANSPORT – DIFFUSION & OSMOSIS

KEY VOCABULARY

Diffusion: net movement of particles from high to low concentration.

Osmosis: diffusion of water across a partially permeable membrane.

Active transport: moving substances against the gradient using energy.

Concentration gradient: the difference in concentration between two areas.

COMMON MISCONCEPTIONS

~~— Diffusion needs energy from respiration.~~

✓ Diffusion is passive — it needs no energy.

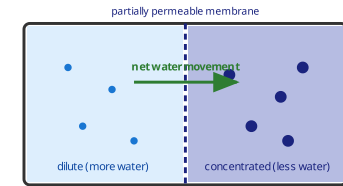
~~— Osmosis is the movement of any substance.~~

✓ Osmosis is the movement of water only.

~~— Particles stop moving once balanced.~~

✓ Particles still move, but there is no net change.

OSMOSIS ACROSS A MEMBRANE



Water moves from a **dilute** solution to a **concentrated** one.

1 DEFINE DIFFUSION

Describe what is meant by **diffusion**. [2 marks]

2 OSMOSIS

Explain how osmosis is different from diffusion. [2 marks]

3 RATE OF DIFFUSION

State **three factors** that increase the rate of diffusion. [3 marks]

4 REQUIRED PRACTICAL

Potato cylinders are placed in sugar solution and lose mass. **Explain why**. [2 marks]

5 PERCENTAGE CHANGE

A cylinder changes from 5.0 g to 4.0 g. **Calculate** the percentage change in mass. [2 marks]

Answer: _____ %

6 ACTIVE TRANSPORT

Complete the sentence using the word bank. [2 marks]

Active transport moves substances _____ the concentration gradient and requires _____.

against

energy

with

no energy
